

MATRA-X — Technical Report

SIH2026 · SIH26099

AI-Driven Standardization and Harmonization of Material Codes Across CPSEs

Team: Alchemy Architects

Live application: <https://matra-x-vision-core.base44.app/>

Abstract

MATRA-X is an enterprise-oriented material intelligence and governance platform designed for heterogeneous CPSE material master data. The current product combines Command Center analytics, AI Standardization, Error Detection, Biometric Authentication activity, Material Cataloguing, Cross-CPSE Insights, LIORA and event/scan activity.

The engineering direction extends the product into an evidence-aware, contradiction-aware and human-governed material identity layer. Structured Engineering DNA is combined with semantic retrieval and technical constraints so that high text similarity cannot silently override a critical engineering conflict.

1. SIH26099 alignment

The solution is designed around material matching, duplicate/near-duplicate/equivalence discovery, standardization, proposed common-code mapping, CPSE mapping, legacy rationalization, human validation, analytics, auditability and ERP/SAP-ready integration.

2. Current application inventory

Command Center; AI Standardization; Error Detection; Biometric Authentication; Material Catalog; Cross-CPSE Insights; LIORA; Event/Scan Activity.

3. Current connected snapshot

Snapshot reviewed 2026-10-02.

MaterialCode: 289 records returned.

Status: standardized 250; flagged 15; error 4; none/unspecified 20.

CPSE: BHEL 40; NTPC 38; ONGC 37; SAIL 37; HAL 36; GAIL 34; IOC 34; BEL 33.

Error type: none 250; duplicate 7; obsolete 2; non_standard 4; anomaly 4; mismatch 2; missing/unspecified 20.

AuthLog: 10 records; face 5; iris 3; fingerprint 2.

ScanEvent: 14 records; biometric 7; standardize 4; verify 1; error_check 2. Severity: info 10; warning 3; critical 1.

These figures are observations from the connected prototype environment. They must not be presented as confidential CPSE production statistics unless their provenance is independently established.

4. Engineering DNA 2.0

Represent each material using structured engineering identity: type, function, subtype, category, family, dimensions, size, material grade, ratings, standards, UOM, manufacturer/part number, application, operating conditions, criticality, aliases, source records, evidence, confidence and uncertainty.

5. Hybrid matching

Ingest → normalize → schema harmonize → Engineering DNA → candidate retrieval → hybrid matching → contradiction guard → uncertainty → explanation → human review → canonical identity → legacy mapping → prevention/analytics.

Signals include lexical similarity, fuzzy similarity, semantic embeddings, vector retrieval, structured attribute compatibility, category compatibility, criticality and evidence quality.

6. False-Merge Firewall

Semantic similarity is candidate evidence, not technical proof. A pair such as HEX BOLT M16 X 80 SS304 versus HEX BOLT M16 X 80 SS316 may be textually similar, but the critical grade conflict should trigger TECHNICAL CONFLICT and HUMAN REVIEW rather than unsafe consolidation.

7. Error detection and Code Doctor

Detect duplicate codes, malformed descriptions, missing critical fields, unit inconsistency, classification mismatch, obsolete mappings and cross-field contradictions. Code Doctor can inspect syntax, uniqueness, source ownership, category consistency, canonical identity and legacy mapping.

8. Evidence fusion

Reconcile permitted master records, technical datasheets, certificates, drawings and manufacturer information. Conflicting high-quality sources should reduce certainty and create a review task.

9. Knowledge graph

Model Material, CPSE, Category, Family, Standard, Supplier, Document, Attribute, Canonical Identity, Legacy Code, Reviewer, Decision, Procurement and Inventory nodes with relationships such as equivalent_to, duplicate_of, near_duplicate, substitute_for, incompatible_with, specified_by, mapped_to, reviewed_by and supported_by.

10. LIORA

LIORA is the MATRA-X Material Intelligence Copilot. It should search materials, compare records, explain decisions, diagnose errors, explain Engineering DNA, inspect evidence, explain legacy mappings, answer SIH26099 questions and query procurement/inventory intelligence.

Grounding pattern: question → intent → MATRA tool → retrieval → evidence → reasoning → answer.

LIORA must not fabricate CPSE records, official common-code adoption, financial savings or engineering approvals.

11. Security and governance

Use role-based access, secure server-side secrets, source/tenant separation, audit events, model/rule versioning and WebAuthn/passkeys for high-risk actions. Raw fingerprint, face and iris images/templates must not be stored by MATRA-X.

12. Scalability

Use indexed databases, vector indexes, candidate generation, asynchronous ingestion, caching, pagination and table virtualization. Million-scale capacity should be demonstrated as a labelled simulation unless actual permitted production data exist.

13. Research validation

Compare normalized/exact, fuzzy/lexical, embedding-only, embedding-plus-attributes and MATRA-X hybrid approaches. Measure precision, recall, F1, top-K recall, false merge rate, critical-conflict recall, calibration, review rate, latency and throughput.

14. Deployment

The evaluator-facing system must work from a public HTTPS URL without a local command prompt. Current live application: <https://matra-x-vision-core.base44.app/>

15. Conclusion

MATRA-X is best presented as an AI-assisted material identity, reasoning and governance layer rather than a simple duplicate detector.

****One material. One governed identity.****